DATA ITEM DESCRIPTION

Form Approved
OMB No 0704-0188

1. TITLE	2. IDENTIFICATION NUMBER
CHEMICAL AND PHYSICAL PROPERTIES FOR FORGING OR CASTING ANALYSIS REPORT	DI-FORG-81196

3. DESCRIPTION/PURPOSE

3.1 The Chemical and Physical Properties for Forging or Casting Analysis Report provides information on the chemical composition and mechanical properties of forgings or castings.

4. APPROVAL DATE (YYMMDD)	5. OFFICE OF PRIMARY RESPONSIBILITY (OPR)	6a. DTIC APPLICABLE	6b. GIDEP APPLICABLE
910502	MR		

7. APPLICATION/INTERRELATIONSHIP

7.1 This Data Item Description (DID) satisfies the requirements of paragraph 3.12 of MIL-S-46172A.

7.2 This Data Item Description (DID) supersedes DI-P-1636.

8. APPROVAL LIMITATION	9a. APPLICABLE FORMS	9b. AMSC NUMBER
		A6120

10. PREPARATION INSTRUCTIONS:

10.1 Reference documents. The applicable issue of the documents cited herein, including their approval dates and dates of any applicable amendments, notices, and revisions, shall be as specified in the contract.

10.2 Format. The Chemical and Physical Properties for Forging or Casting Analysis Report format shall be contractor selected. Unless effective presentation would be degraded the initially used format arrangement shall be used for all subsequent submissions.

10.3 Content. The Chemical and Physical Properties for Forging or Casting Analysis Report content shall be in accordance with the elements of Figure 1 and the following requirements.

10.3.1 General information. RECORD OF TREATMENT AND TEST OF (enter appropriate identification), Contract (name and address of contractor), Contract Order No., Date, Specification No., Drawing No., Forging/Casting No., Heat No., Date Cast/Forged, Shipped To (Released), and Date Shipped, Released.

10.3.2 Chemical analysis. The complete Obtained and Required chemical composition (C - carbon, MN - manganese, P - phosphorus, S - sulfur, Si - silicon, Ni - Nickel, Cr - chromium, Mo - molybdenum, V - vanadium, blank - other) of the forging or casting.

(Continued on Page 2)

11. DISTRIBUTION STATEMENT

DISTRIBUTION STATEMENT A: Approved for public release; distribution is unlimited.

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10. PREPARATION INSTRUCTIONS (Continued)

10.3.3 Physical information. Macroetch Satisfactory, Macroetch Not Taken, Cold Worked and Soaked, Not Cold Worked, Ingot No., Type of Ingot, Ingot Size, Length (inches), Total Weight, Wt (weight) Above Piece, Wt (weight) Below Piece, Date Forged.

10.3.4 Heat treatment procedure. A complete and detailed description of the heat treatment procedure including the date completed.

10.3.5 Mechanical properties. Specimen Locations, Y.S. (yield strength), T.S. (tensile strength), Elon Percent (percent elongation), R.A. Percent (reduction of area %), Hardness, Fracture, and Charpy (Ft-Lb) values.

10.3.6 Authentication. The Chemical and Physical Properties for Forging or Casting Analysis Report shall include identification of the person performing the physical tests (PHYSICAL TESTS MADE BY) and shall be signed by the appropriate Government Inspector with the Place and Date identified.

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Chemical and Physical Properties for Forging or Casting Analysis

RECORD OF TREATMENT AND TEST OF[1] _____ WATERVLIET ARSENAL

Contract[2]								Contract Order No			Date
Specification No				Drawing No				Forging Casting No			
Heat No		Date Cast			Shipped To					Date Shipped, Released	
Chemical Analyses Made By:											Macroetch: Satisfactory
	C	Mn	P	S	Si	Ni	Cr	Mo	V		Macroetch: Not Taken
Obtained											Cold Worked and Sealed
Required											Not Cold Worked
Ingot No		Type of Ingot						Ingot Size			Inches Length Inches
Total Weight		Wt Above Piece						Wt Below Piece			Date Forged

TREATMENT	PHYSICAL TESTS MADE BY						
Date Completed:	Specimen Locations	Y.S.	T.S.	Elon Percent	R.A. Percent	Hardness[3]	Fracture[4]
	REQUIRED						

Specimen Locations	Charpy (Ft-Lb)	Remarks:

1 Enter the numbers of ALL forging or castings represented by these test; marking with an asterisk those from which specimens are taken.
2 Enter firm name and address.
3 Type of hardness machine.
4 C-Capped Cr-Crystalline I-Irregular N-Nonmetallic
P-Pitted L-Laminated W-Woody L-Lustrous Cavities

Place	Date
Government Inspector	

Figure 1. Sample Format

DI-FNEC-81194

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